

Cyberdeck Cellular & Media

The full written lesson — LTE modems & AT commands, APN & carrier auth, SMS & voice, Linux audio & power cost.

1. LTE modems and AT commands

A SIM7600-class LTE HAT doesn't show up as one clean network device — it shows up as several USB serial ports at once, and only one of them speaks the language you actually control it with.

- Plugging one in typically creates several `/dev/ttyUSB0` – `ttyUSB3` -style devices: AT commands, GPS NMEA output, and the actual PPP/data connection each get their own.
- **AT commands** are short text commands the modem's firmware understands — `AT` checks it's alive, `AT+CSQ` reports signal quality, `AT+CREG?` reports network registration, `AT+COPS?` reports the current carrier.
- **ModemManager** (via `mmcli`) is the higher-level Linux service most desktops use. Talking to the AT port directly with `minicom` or `screen` gives more direct control and is easier to debug.

The trap: sending AT commands to the wrong serial port. Only one of the modem's several `/dev/ttyUSBx` devices accepts AT commands — sending `AT` to the wrong one looks exactly like "the modem is dead."

Worked: `AT+CREG?` replies `+CREG: 0,1`. The second number: `1` = registered home, `5` = registered roaming, `0` = not registered. A stuck `0,0` is almost always a band/antenna/SIM problem.

2. APN configuration and carrier authentication

Being registered on the network and having a working data connection are two completely separate achievements.

- **Registration** means the modem authenticated to the carrier's radio network — voice/SMS can already work. **Data** additionally requires a correctly configured **APN**.
- APNs are carrier-specific strings, set via `AT+CGDCONT=1,"IP","<apn>"`.
- SIM7600-class modules generally need **both a main and a diversity/MIMO antenna** for reliable LTE — one alone can still register, just with degraded throughput.

The trap: "registered but no data" is a diagnostic fork. No registration → band/antenna/SIM. Registered but no data → APN, first. Confusing which symptom you have wastes time debugging the wrong layer.

Worked: registration confirmed, but `ping` times out. Check `AT+CGDCONT?`, compare against the carrier's published APN, correct it, re-test — before assuming a hardware fault.

3. SMS and voice calls through the modem

Text messaging and dialing a call both reduce to a short AT command sequence — each with one easy-to-miss detail.

- **SMS text mode:** `AT+CMGF=1` switches from the default **PDU mode** (hex encoding) to plain text. Set this before anything else SMS-related.
- **Sending:** `AT+CMGS="+1..."`, then the message body, ended with `Ctrl+Z`.
- **Voice:** `ATD+1...;` dials — the trailing **semicolon** marks it as voice, not data. `ATA` answers, `AT+CHUP` hangs up.

The trap: following a walkthrough that assumes PDU mode without saying so. `AT+CMGS` in PDU mode expects a completely different format — always confirm `AT+CMGF=1` first.

Worked: a call connects (`ATD...;` returns `OK`) but there's no audio either direction. That's a *separate* problem from the AT sequence — audio routing needs its own configuration.

4. Linux audio and what cellular costs the battery

Sound and power draw are unrelated on paper, but both show up as "it mostly works but something's subtly wrong" problems.

- **ALSA** is the low-level Linux audio layer. **PulseAudio/PipeWire** sits on top to mix multiple simultaneous streams — call audio and music playback both need that layer.
- **Ground-loop noise** from a cheap audio interface sharing a ground with an actively-transmitting modem is an *electrical* problem — no software config fixes it.
- **Transmit bursts draw far more current than idle registration** — a simple average-current measurement can understate the real peak.

The trap: sizing a power supply for idle average current and being surprised by resets specifically during SMS sends, calls, or uploads — exactly when current spikes hardest.

Worked: ~200mA idle-registered, spiking to 1.5–2A during a transmit burst. Sizing only for the average risks a brownout — and a full reset — during that burst.